

Plastic pollution has also become increasingly recognized as a global challenge as millions of tonnes are added to oceans every year, and some of which will persist for long periods of time. The accumulation of plastics harms by threatening marine ecosystems, human health, and biodiversity.

Traditional methods of detection and monitoring may be financially and practically challenging (if you want to monitor continuously or in large scales). However, there are remote sensing alternatives such as satellite imagery and UAVs that offer potential new viable and scalable solutions for plastic detection, including through their unique spectral signatures, especially when reflectance is considered in the Near-Infrared.

Advances in ML and DL have enhanced the potential of remote sensing by automating analysis of complex spectral and spatial patterns. Indices like Floating Debris Index(FDI) and Plastic Index(PI) have differentiated between plastics and natural marine features.

The current research highlights the need for reliable detection but also quantification of plastic mass fluxes, identification of pollution hotspots, and development of a scalable, long-term monitoring framework.

5. Related Work

Sr. No	Name of the Study	Features	Methodology	Research Gaps Present
1	Detection of marine floating plastic using Sentinel-2 imagery and machine learning models [1]	This study was successful in detecting plastic across Greece, Cyprus, Italy, and Lebanon. It noted different spectral responses, plastic bottles reflecting high Near Infrared (NIR) reflectance, and achieved ~99% model accuracy for certain sites. The Floating Debris Index (FDI) was noted to be the most important variable.	The methodology used data from Sentinel satellites and Support Vector Machines (SVMs) and Random Forest (RF) classification models. Models were trained with in situ plastic location data and remote sensing bands and indices (FDI, NDVI, PI, kNDVI). The images were atmospherically corrected using ACOLITE and C2RCC processors.	There is a need for high-quality training data and a comprehensive spectral library. A requirement to mitigate false alarms from phenomena like sea plume was noted. A broader assessment of advanced deep learning (DL) algorithms (e.g., DeepLab, YOLO) is needed.

2	High-precision Density Mapping of Marine Debris and Floating Plastics via Satellite Imagery [2]	The study developed MAP-Mapper models with high precision (up to 95%) and introduced the Marine Debris Mapping (MDM) index to quantify density. It demonstrated the ability to detect plastics at a sub-pixel scale and mapped highly polluted regions in the Gulf of Honduras, Manila, and Mumbai.	The work combined Sentinel-2 data with U-net-based ML models, using the MARIDA and PLP datasets for training and validation. An automated data pipeline (MAP-Mapper) was developed, and atmospheric correction was performed with ACOLITE. Only four Sentinel-2 bands were used to improve efficiency.	Future work should expand the MARIDA dataset. A more rigorous validation of density mapping using ground truth data is required. Misclassifications of small boats, waves, and sea foam need to be addressed, and the use of minimal-supervision learning approaches is promoted.
3	Deep learning for detecting macroplastic litter in water bodies: A review [3]	This is a systematic review of 34 DL-based studies. It identified key gaps, including a lack of model generalisation and the absence of mass flux quantification. Most studies focus on marine environments, neglecting rivers and lakes. Object detection is the most common computer vision task.	A systematic review methodology was used to analyse 34 papers from "Scopus" and "Web of Science" databases up to 2021. Papers were categorised by water body, dataset details, computer vision task, and performance metrics.	A primary need exists for DL models with robust generalisation across different conditions. There is a lack of methods for quantifying macroplastic mass fluxes and hotspots. Insufficient research on riverine pollution and inconsistent reporting of metrics across studies were also identified as gaps.

4	Litter Detection with Deep Learning: A Comparative Study [4]	This paper provided a comparison of existing DL object detection models and a new dataset, "PlastOPol". The best accuracy from the existing models was YOLO-v5x, but the best potential for mobile devices was YOLO-v5s. All the object detection methods reported difficulty in detecting low sizes of litter and cluttered or complex backgrounds.	The study considered models based on CNNs such as the Faster R-CNN, Mask R-CNN, EfficientDet, RetinaNet, and YOLO-v5. Performance was evaluated by effectiveness (AP50) and efficiency (model size, FPS) when evaluated on the PlastOPol and TACO datasets with 5-fold cross-validation.	Future work should enhance detection using fusion strategies and continue optimising tiny neural networks for constrained devices. There is a need to improve datasets with smart data augmentation and add multiclass annotations to the PlastOPol dataset.
5	Review of Methods for Automatic Plastic Detection in Water Areas Using Satellite Images and Machine Learning [5]	This comprehensive review highlights Sentinel-2 as the primary satellite for plastic detection. It notes FDI and PI as key spectral indices and finds that supervised algorithms (SVM, RF, XGBoost) are generally more efficient.	Information from over 50 studies was synthesised. The review discussed data acquisition (satellites, UAVs), image processing, spectral indices, and a wide range of ML/DL algorithms.	There is a demand to collect global plastic pollution research data where AI can be trained, this would also facilitate the creation of global database of types of plastic litter. The paper indicates that it would be worth exploring other satellites for further ocean coverage and should also consider the opportunity for synthetic aperture radar (SAR) imagery.
6	Ocean pollution detection using image processing [6]	The study proposed an economical method for predicting plastic content in water, achieving 80% to 90% accuracy. The method enables automated and continuous monitoring of pollution events.	The methodology applied image processing with a Convolutional Neural Network (CNN). The process involved color correction, histogram equalisation, and feature extraction using Discrete Wavelet Transform (DWT) and Gray Level Co-occurrence Matrix (GLCM).	The main mentioned limitation is that image processing algorithms can suffer from varying accuracy. Gaps we inferred included issues with generalisability and robustness under varying real world environmental conditions (variations in light and complex water surface).

6. Methodology

The methodology leveraged both basic and advanced digital image processing tools, mostly for the overall data preparation and features extraction:

Data Acquisition and Preprocessing.

- **Dataset:** The study utilized the Trash_can dataset, which comprised 6000 training images and 1200 images for validation.
- **Atmospheric and Noise Correction:** Raw images are often put through processing steps, as shown in the Atmospheric Correction (eg. ACOLITE Tool) and Masking (land and cloud removal for cleaned surface reflectance) steps, to obtain precise surface reflectance data.
- **Image Enhancement:** This process is fundamental to making features usable. All of which can be accomplished through
- **Color correction:** White balancing is a color correction method that corrects color casts like a strong or excessive green tint. The effectiveness of the white balancing technique showed to be useful in eliminating casts without producing other artifacts or extreme amelioration of hues.
- **Contrast enhancement:** Contrast can be improved using histogram equalisation or, more specifically, CLAHE (Contrast Limited Adaptive Histogram Equalization). In the final model training, CLAHE was employed on the L (Lightness) channel after extracting it, considering its narrow histogram verified low contrast in the original image.
- **Simplification of Complexities:** The transition from RGB to gray-scale imagery was intended to lessen the complexity of features that were previously visualized in RGB.
- **Feature Extraction:** Spectral indices were devised within the analysis to differentiate between plastics and natural features in order to classify associated debris. The two primary spectral indices are the Floating Debris Index (FDI) and the Plastic Index (PI). Other methods of image analysis were purposely designed to analyze data utilizing spectral indices, which would analyze the image into frequency images utilizing the Discrete Wavelet Transform (DWT) and the Gray Level Co-occurrence Matrix (GLCM).
- **Model Selection & Training - Architectures:** Object Detection (OD) models were used specifically the YOLO (You Only Look Once) family of models (ie YOLOv5s, YOLOv5x) for the model's speed and its ability to run on real-time data.
- **Model Comparisons:** The evaluation of the analysis also included a performance assessment against a pre-processed baseline model that was trained from images that were not processed, which were collected in their original form for analysis.

7. Results

The specific image processing preprocessing steps were notably successful (CLAHE and white balancing) with both quantifiable and qualitative results, as below will describe:

Visual and Qualitative Results

- The preprocessing steps delivered an improved visual dataset.
- CLAHE successfully produced clearer and higher-contrast images with an enhancement in local detail.
- White balancing created more natural images, producing better color fidelity by correctly neutralizing a strong green tint that the original data had.

Quantitative Model Performance Results

- The improved dataset produced a better model.
- The preprocessing pipeline was very successful with an important jump in segmentation mAP score from 0.1815 to 0.2955.
- The model gained a significant amount of accuracy and reliability from the baseline, as below:
 - A precision improvement from 0.46 to 0.63.
 - A recall improvement from 0.37 to 0.54.

(To give context, the baseline model trained without preprocessing had a Box mAP50-95 of 0.36287, and a Mask mAP50-95 for 0.29554).

8. Conclusion

The project effectively illustrated the important role of certain digital image processing techniques to enhance deep learning performance for environmental monitoring.

- When procedural steps were combined, (CLAHE for contrast, and white balancing for color balance), it not only provided a visually superior dataset, but also a more effective model.
- The use of CLAHE to improve local contrast and white balancing to neutralize color casts were successful methods for clarifying details and making the images seem more natural.
- The pipeline in which the method was applied produced considerable improvements in the most pertinent operational metrics of the analysis, especially increases in segmentation mAP, precision and recall, thus confirming improvements in the accuracy and reliability of the model.
- The investigation confirmed that combining high-resolution data (like Sentinel-2 imagery) with advanced image processing and either DL or ML techniques is a legitimate approaches to environmental monitoring. Preparatory steps, including atmospheric correction and masking, are rigorous but critical methods to obtain accurate reflectance data.
- Lightweight DL architectures, specifically YOLOv5s, demonstrate potential for real-time detection capabilities as they can be changed to operate on mobile and edge devices, which represents the feasibility of a practical and scalable implementation of the image processing methods and techniques to environmental monitoring research.

9. References

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